Muscles of thoracic wall and intercostal spaces, its nerves and vessels.



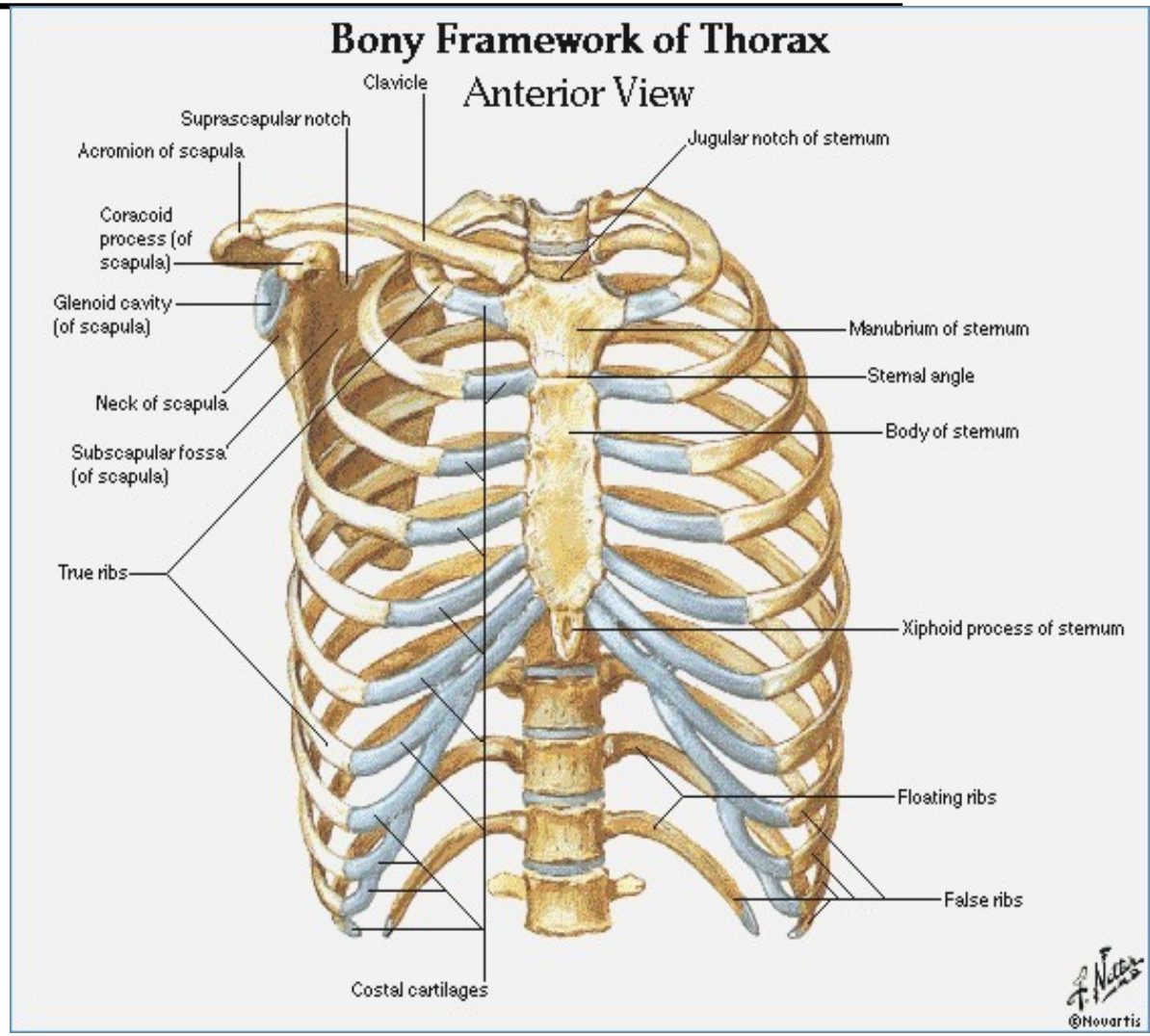
LEARNING OBJECTIVES

- At the end of the lecture the student should be able to know about:
- Muscles of thoracic wall
- Intercostal muscles
- Content of intercostal spaces
- Origin of intercostal arteries
- Origin, course and distribution of intercostal nerves
- Branches and course of internal thoracic artery



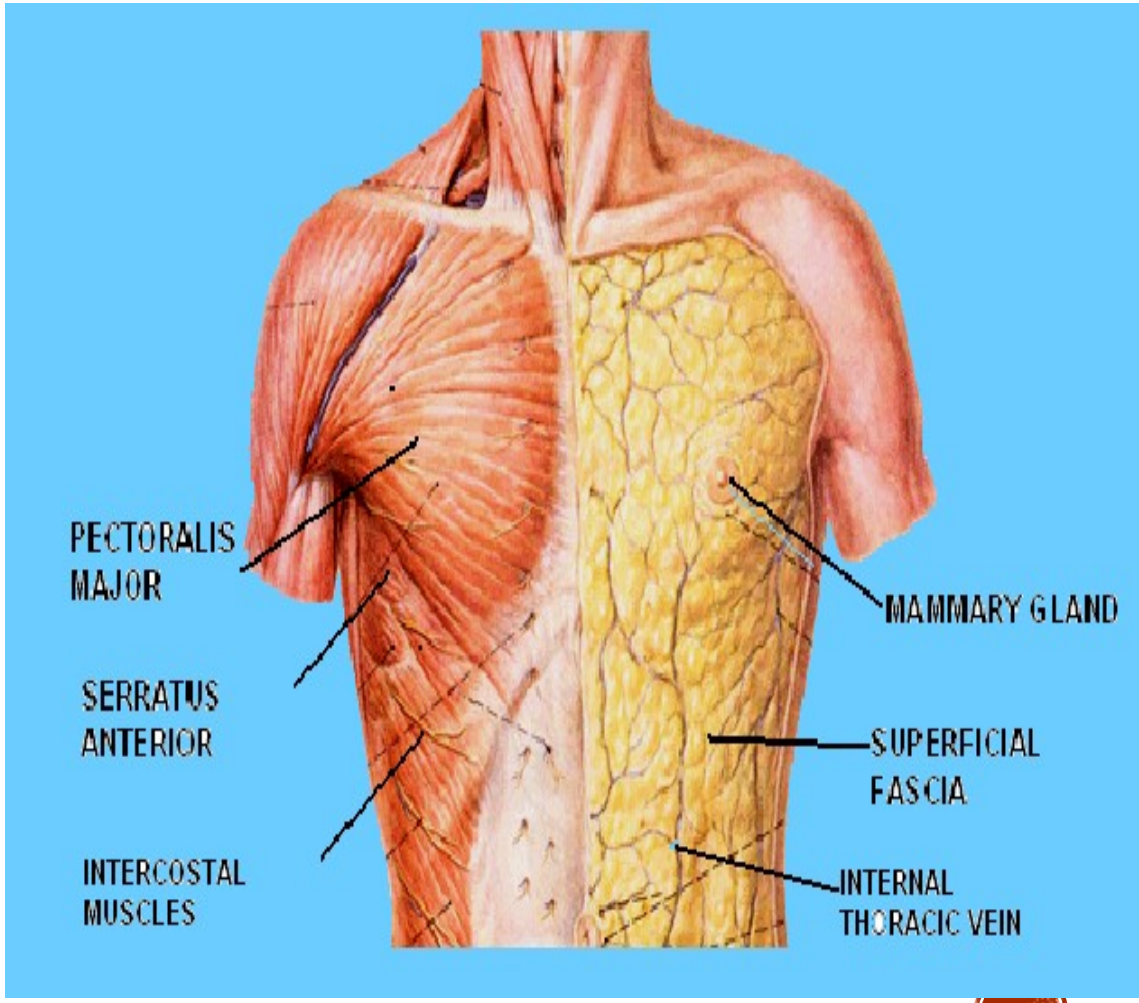
INTRODUCTION TO THORACIC WALL

- The thorax (or chest) is a region of the body between the neck and the abdomen.
- Anteriorly formed by sternum and costal cartilages
- Laterally by ribs and intercostal spaces
- Posteriorly by thoracic vertebrae
- Superiorly by supra-pleural membrane
- Inferiorly by the diaphragm



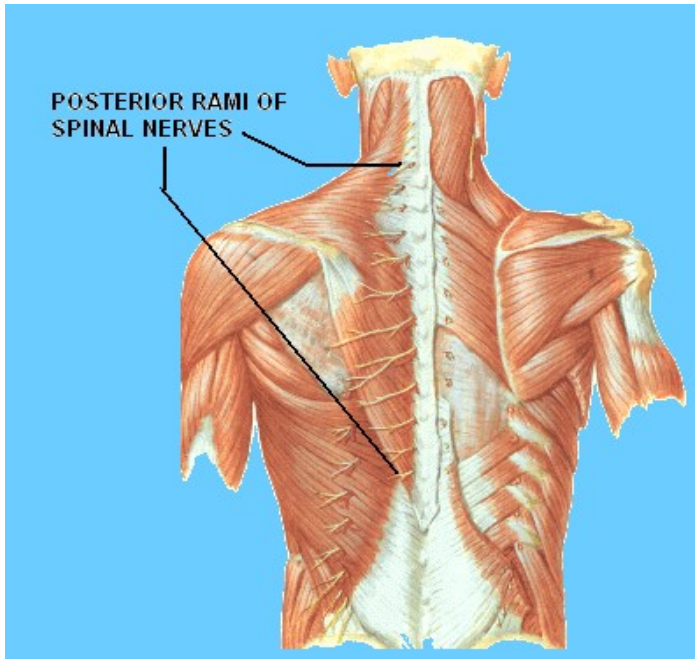
THORACIC WALL

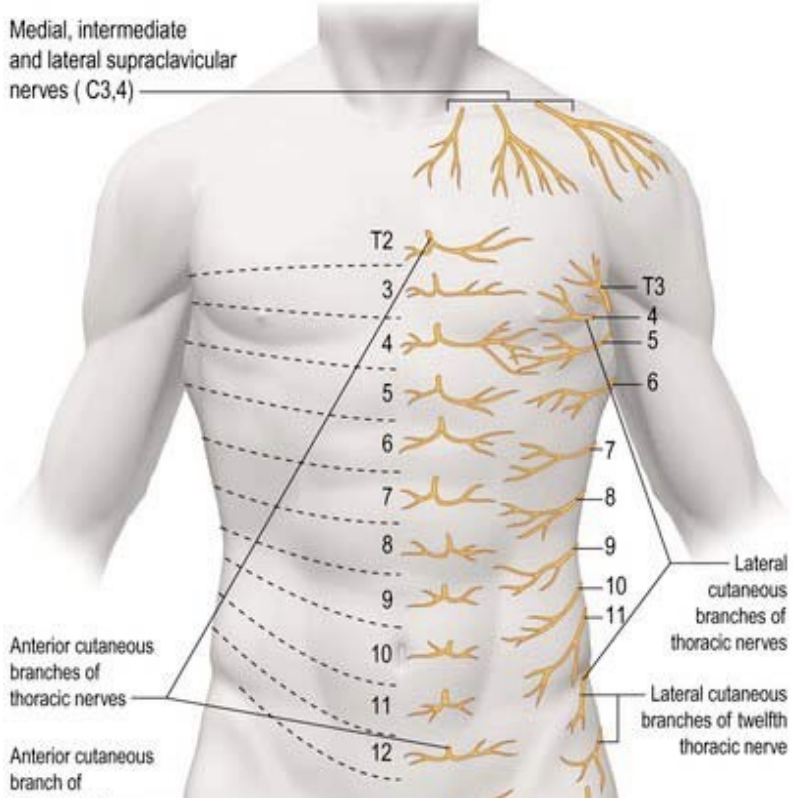
- Covered by the following structures from outside to inside
- Skin
- Subcutaneous tissue
- Superficial fascia
- Mammary gland
- Muscles
- Bones
- Vasculature (vein , artery, nerve)
- Deep fascia– endothoracic fascia
- Parietal pleura



CUTANEOUS INNERVATION OF CHEST WALL

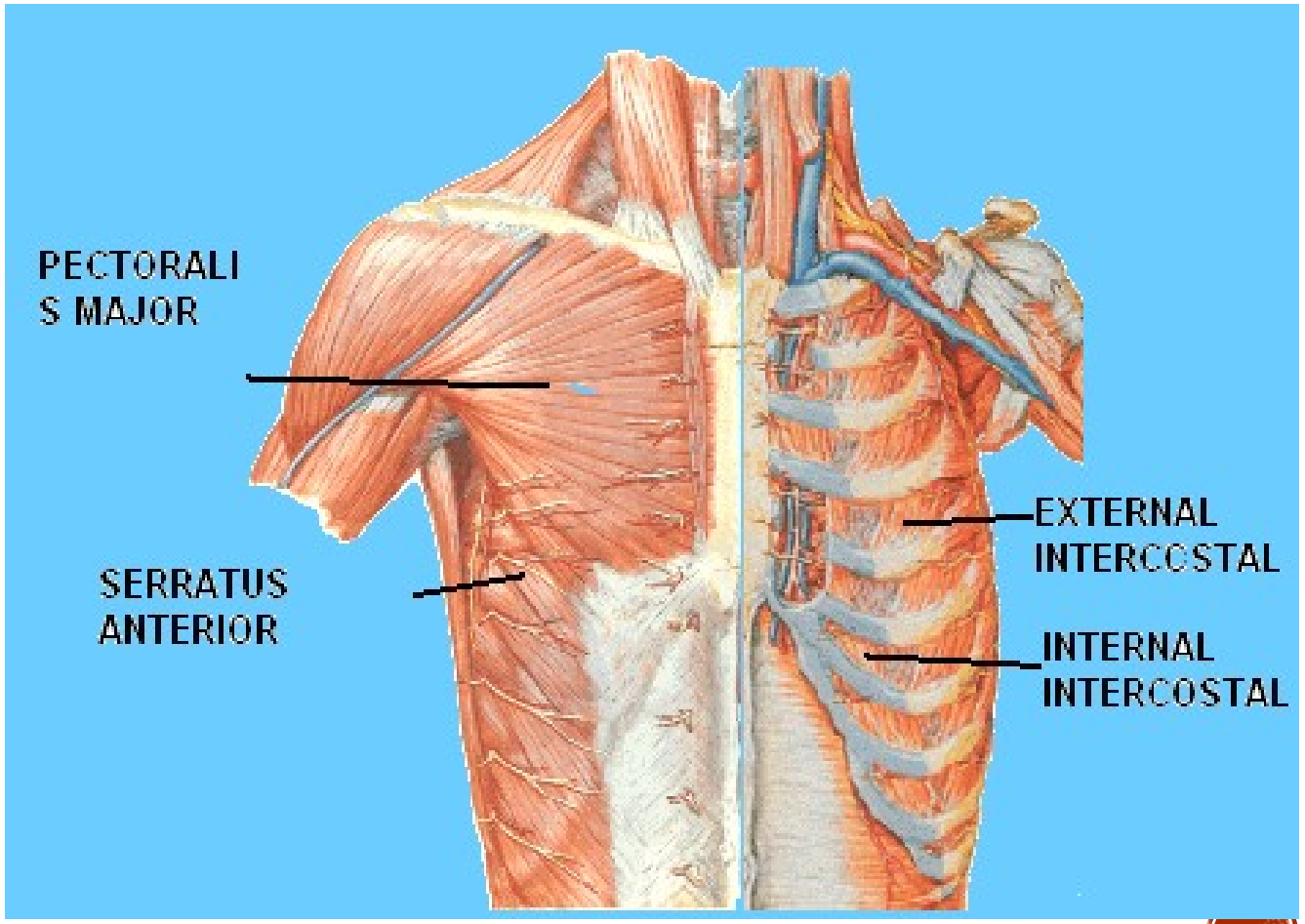
- **Anteriorly**
- •Above sternal angle; from ***Supraclavicular nerves(C3-C4)***
- •Below sternal angle, from ***anterior and lateral cutaneous branches of intercostal nerves***
- **Posteriorly**
- •Dorsal rami of spinal nerves





THE MUSCLES OF THORAX

- **Anterior thoracic wall:**
- **Extrinsic muscles**
- •Pectoralis major
- •Pectoralis minor
- •Serratus anterior
- **Intrinsic muscles**
- External intercostal muscle (11)
- Internal intercostal muscle (11)
- Innermost intercostal muscle



THE MUSCLES OF THORAX

- **Posterior thoracic wall:**
- Intercostal muscles and
- Levatores costarum (12)
- Serratus posterior superior
- Serratus posterior inferior

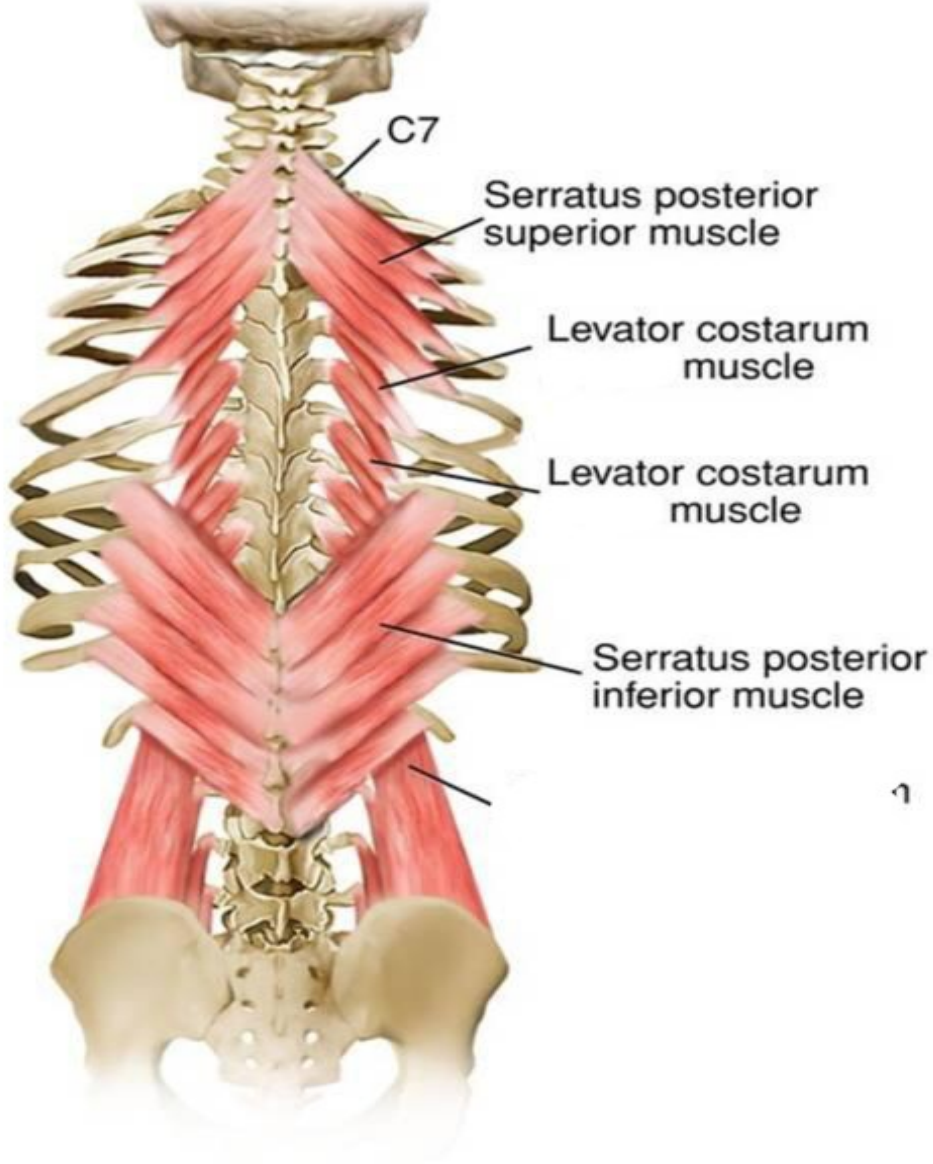


TABLE 9.1 Muscles Connecting the Upper Limb to the Thoracic Wall

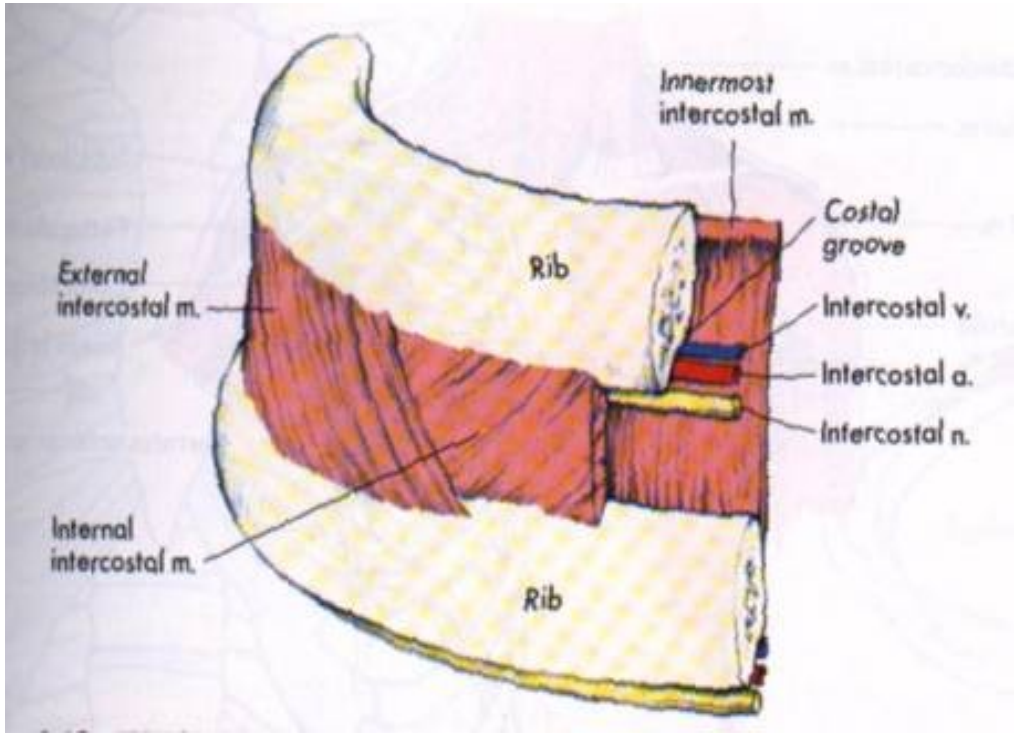
Muscle	Origin	Insertion	Nerve Supply	Nerve Roots[a]	Action
Pectoralis major	Clavicle, sternum, and upper six costal cartilages	Lateral lip of bicipital groove of humerus	Medial and lateral pectoral nerves from brachial plexus	C5, 6, 7, 8; T1	Adducts arm and rotates it medially; clavicular fibers also flex arm
Pectoralis minor	3rd, 4th, and 5th ribs	Coracoid process of scapula	Medial pectoral nerve from brachial plexus	C6, 7, 8	Depresses point of shoulder; if the scapula is fixed, it elevates the ribs of origin
Subclavius	1st costal cartilage	Clavicle	Nerve to subclavius from upper trunk of brachial plexus	C5, 6	Depresses the clavicle and steadies this bone during movements of the shoulder girdle
Serratus anterior	Upper eight ribs	Medial border and inferior angle of scapula	Long thoracic nerve	C5, 6, 7	Draws the forward anterior around the thoracic wall; rotates scapula

TABLE 2.1 Muscles of the Thorax

Name of Muscle	Origin	Insertion	Nerve Supply	
External intercostal muscle (11) (fibers pass downward and forward)	Inferior border of rib	Superior border of rib below	Intercostal nerves	With 1st rib fixed, they raise ribs during inspiration and thus increase anteroposterior and transverse diameters of thorax
Internal intercostal muscle (11) (fibers pass downward and backward)	Inferior border of rib	Superior border of rib below	Intercostal nerves	With last rib fixed by abdominal muscles, they lower ribs during expiration
Innermost intercostal muscle (incomplete layer)	Adjacent ribs	Adjacent ribs	Intercostal nerves	Assists external and internal intercostal muscles
Diaphragm (most important muscle of respiration)	Xiphoid process; lower six costal cartilages, first three lumbar vertebrae	Central tendon	Phrenic nerve	Very important muscle of inspiration; increases vertical diameter of thorax by pulling central tendon downward; assists in raising lower ribs Also used in abdominal straining and weight lifting
Levatores costarum (12)	Tip of transverse process of C7 and T1–11 vertebrae	Rib below	Posterior rami of thoracic spinal nerves	Raises ribs and therefore inspiratory muscles
Serratus posterior superior	Lower cervical and upper thoracic spines	Upper ribs	Intercostal nerves	Raises ribs and therefore inspiratory muscles
Serratus posterior inferior	Upper lumbar and lower thoracic spines	Lower ribs	Intercostal nerves	Depresses ribs and therefore expiratory muscles

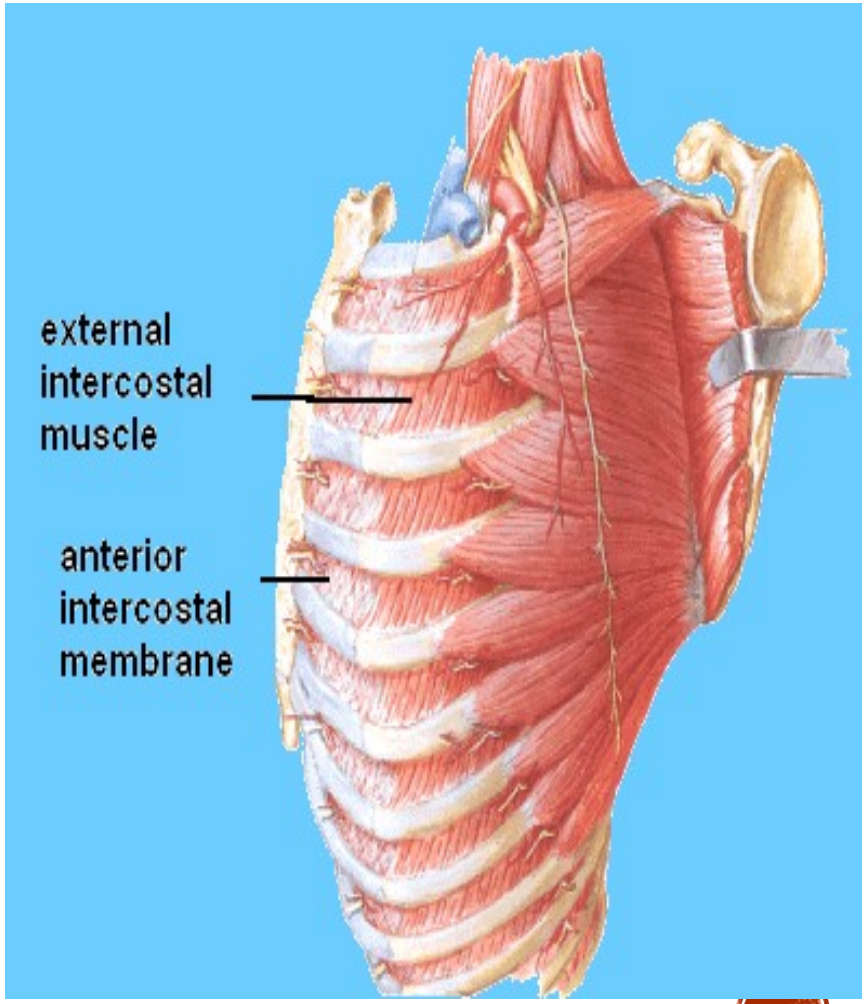
INTERCOSTAL SPACES

- Spaces between ribs
- •11 in number
- •Contain three layers of muscles of respiration
- –External intercostal
- –Internal intercostal
- –Innermost inter costal
- •Last one is lined by endothoracic fascia and then parietal pleura
- •Each intercostal space has its own neurovascular bundle



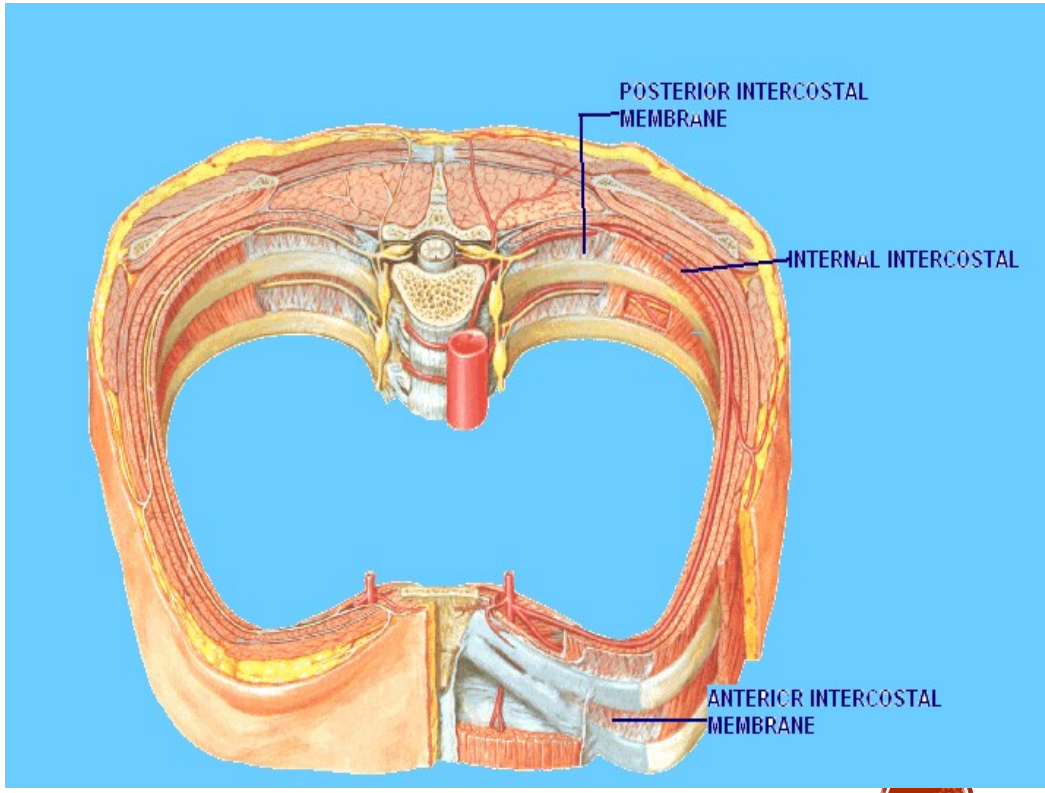
EXTERNAL INTERCOSTAL MUSCLES

- Most superficial of intercostal muscles
- **Origin**
- •Inferior border of the rib above
- •**Insertion**
- •Superior border of the rib below
- **Course**
- •Fibers run downward and forward
- •Extends from the costal tubercle to the costochondral junction where it is replaced by **anterior intercostal membrane**
- •**Action**: elevate ribs adding in forced inspiration



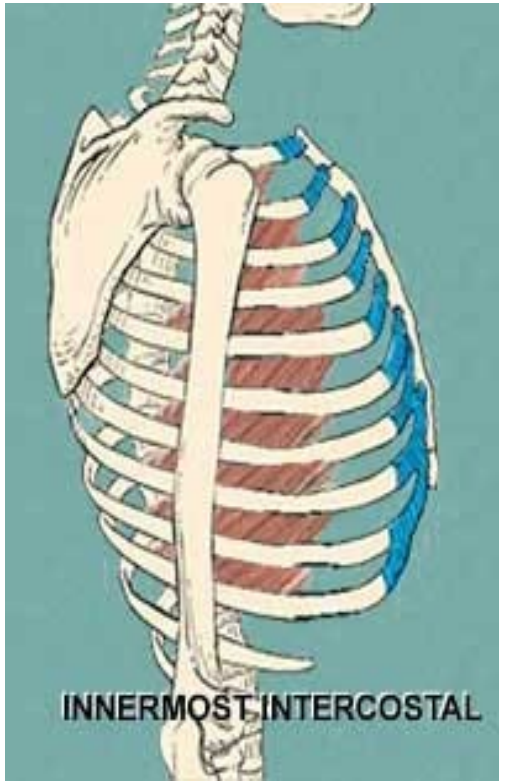
INTERNAL INTERCOSTAL MUSCLES

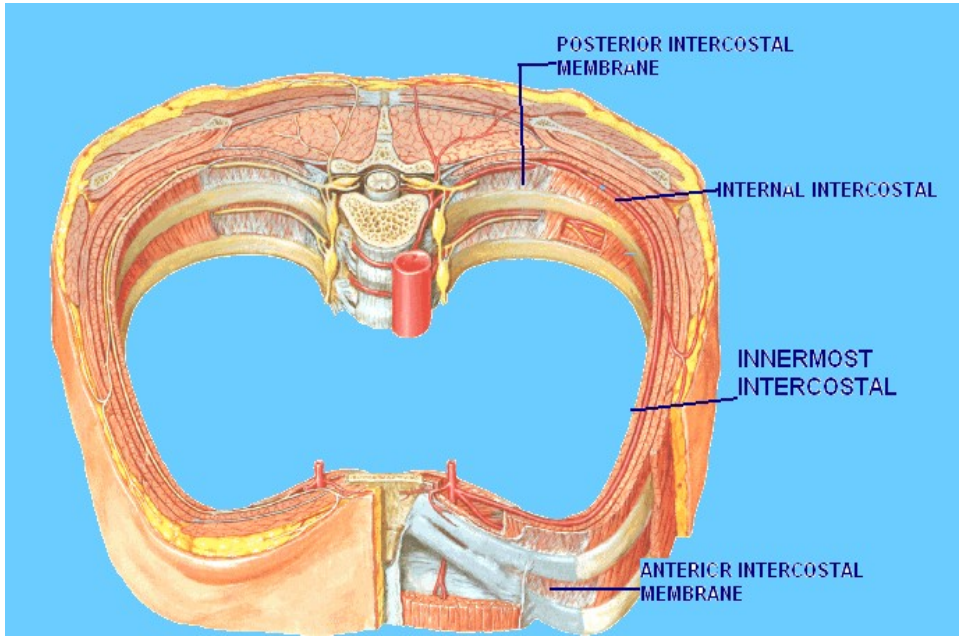
- Forms intermediate layer
- **Origin**
- •Subcostal groove of the rib above
- **Insertion**
- •Upper border of rib below
- **Course**
- •Fibers runs downward and backward from sternum to angle of the rib
- •At angle of the rib it forms ***posterior intercostal membrane***
- •**Action**: depress ribs for forced expiration



INNERMOST INTERCOSTAL

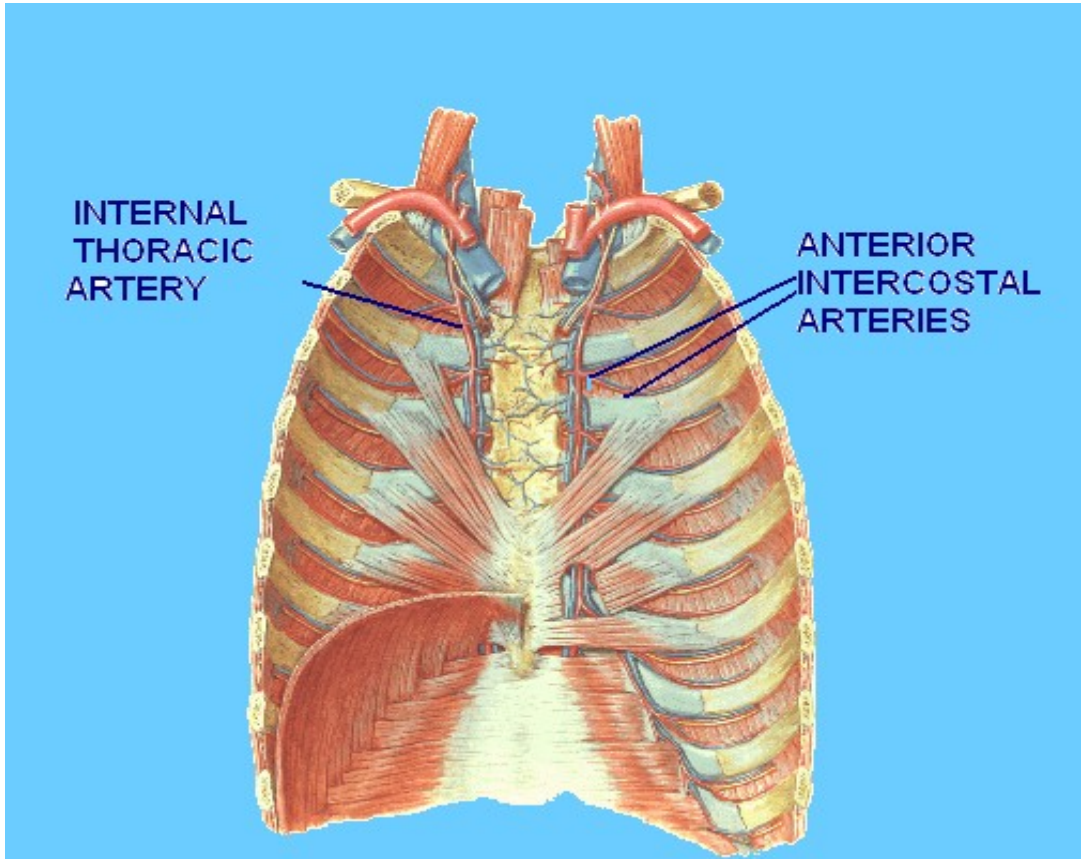
- Deepest layer
- •Crosses more than one intercostal space
- •Related internally to endothoracic fascia and parietal pleura
- •Externally related to intercostal vessels and nerves
- •Assist the intercostal muscles





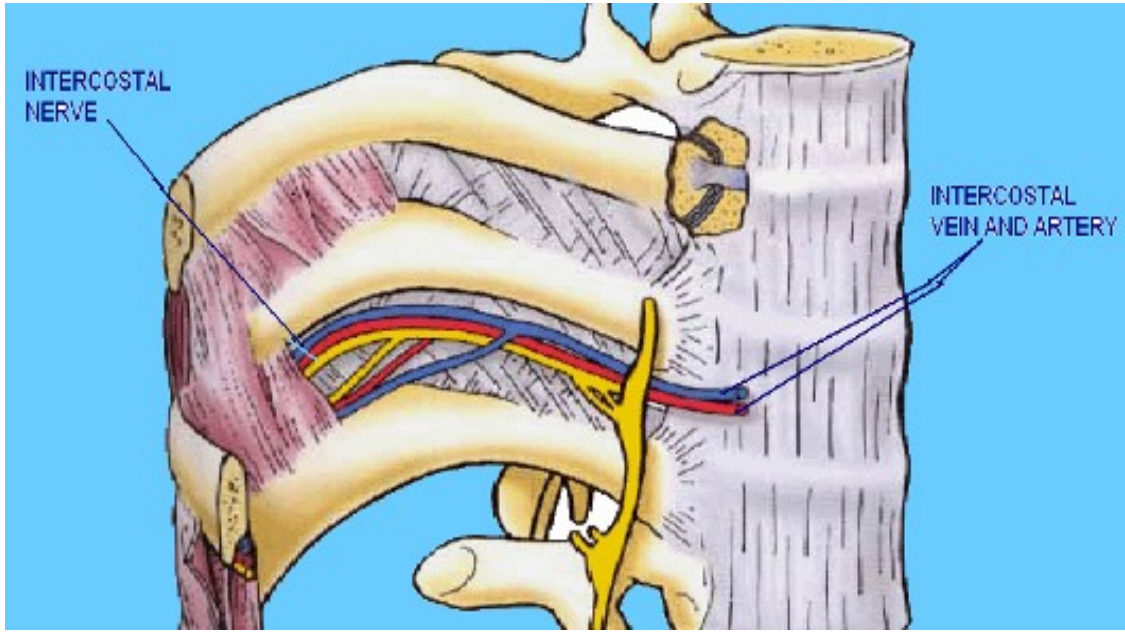
INTERCOSTAL ARTERIES

- Two anterior intercostal and one single posterior intercostal artery in each space
-
- **Anterior intercostal arteries**
- •Of first six spaces from ***internal thoracic artery***
-
- •Of last six are the branches of ***musculophrenic artery***



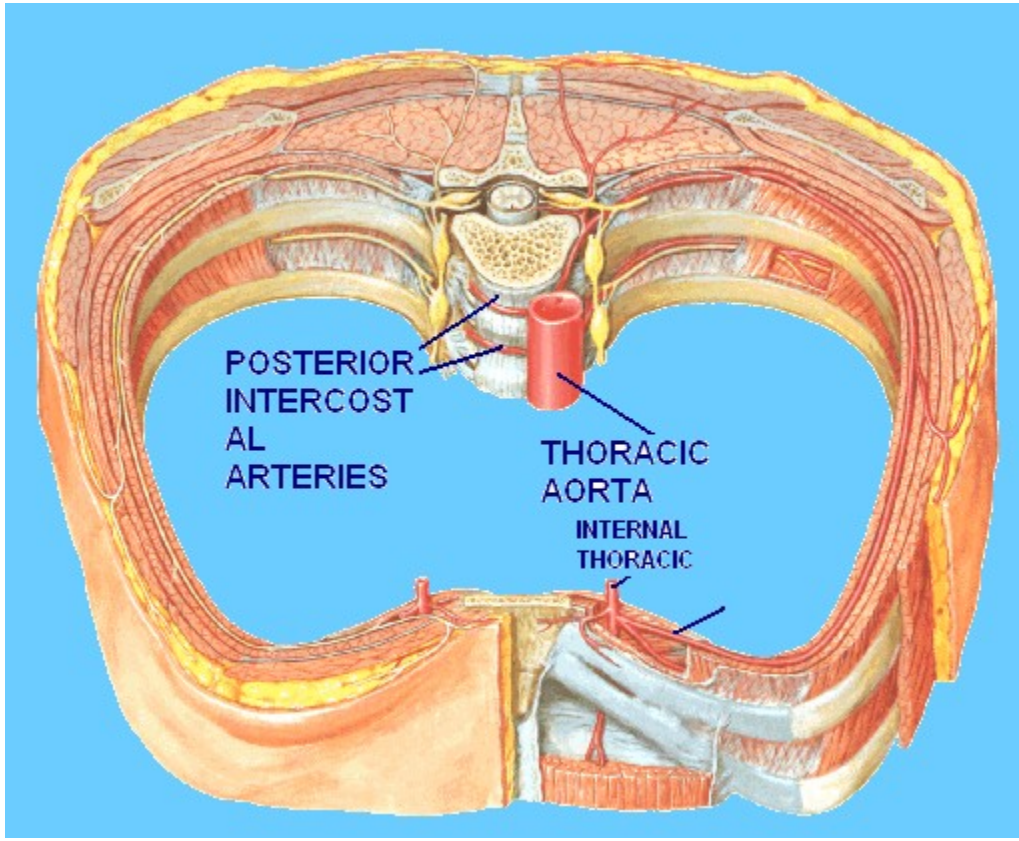
NEUROVASCULAR BUNDLE OF INTERCOSTAL MUSCLES

- Inter costal vessels and nerves
- •Run between internal and innermost intercostal layer
- •Arranged in following order from above downwards
- –**Intercostal vein**
- –**Intercostal artery**
- –**Intercostal nerve**
- •Runs in subcostal groove
- •Runs near the upper margin of concerned intercostal space



INTERCOSTAL ARTERIES

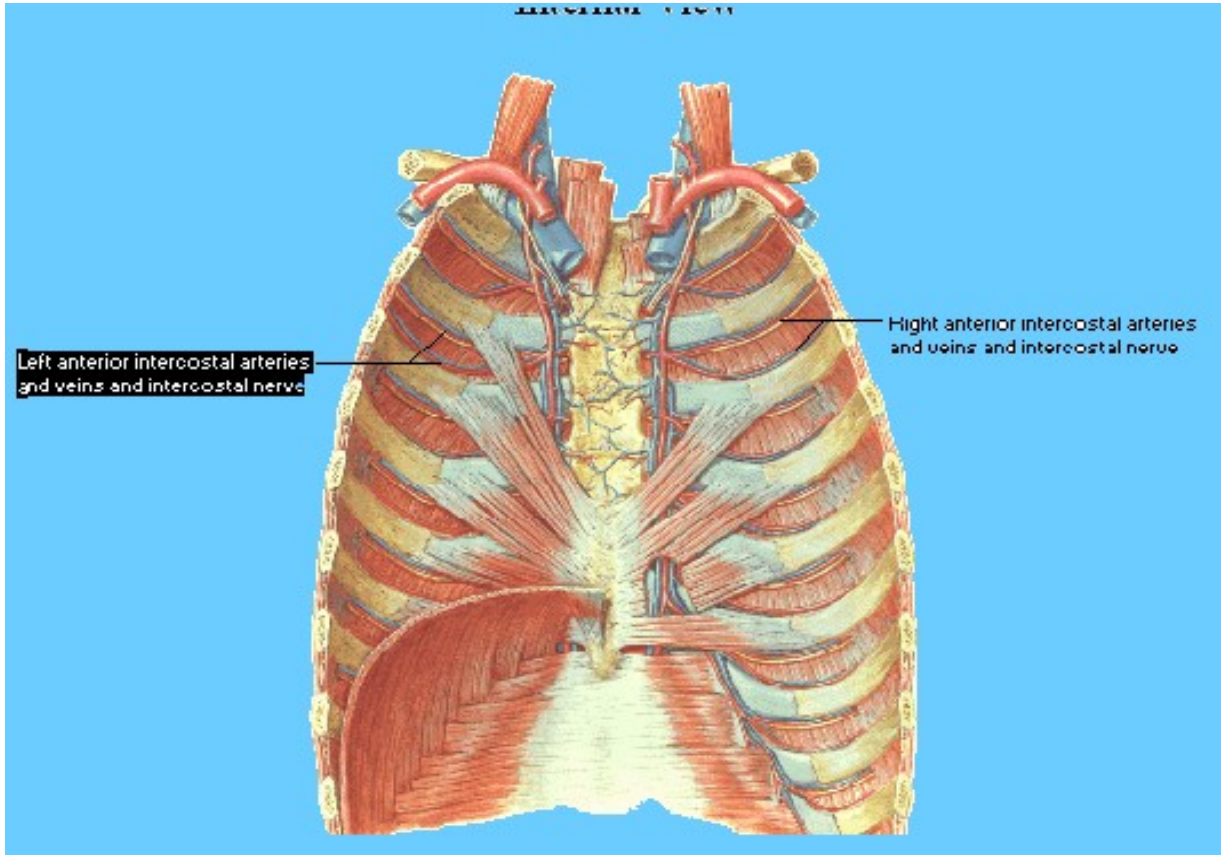
- **POSTERIOR INTERCOSTAL ARTERIES**
- •subclavian artery costocervical trunk **superior intercostal artery** posterior intercostal arteries of first 2 spaces
- •Those of last 9 spaces from **descending thoracic aorta**



INTERCOSTAL VEINS

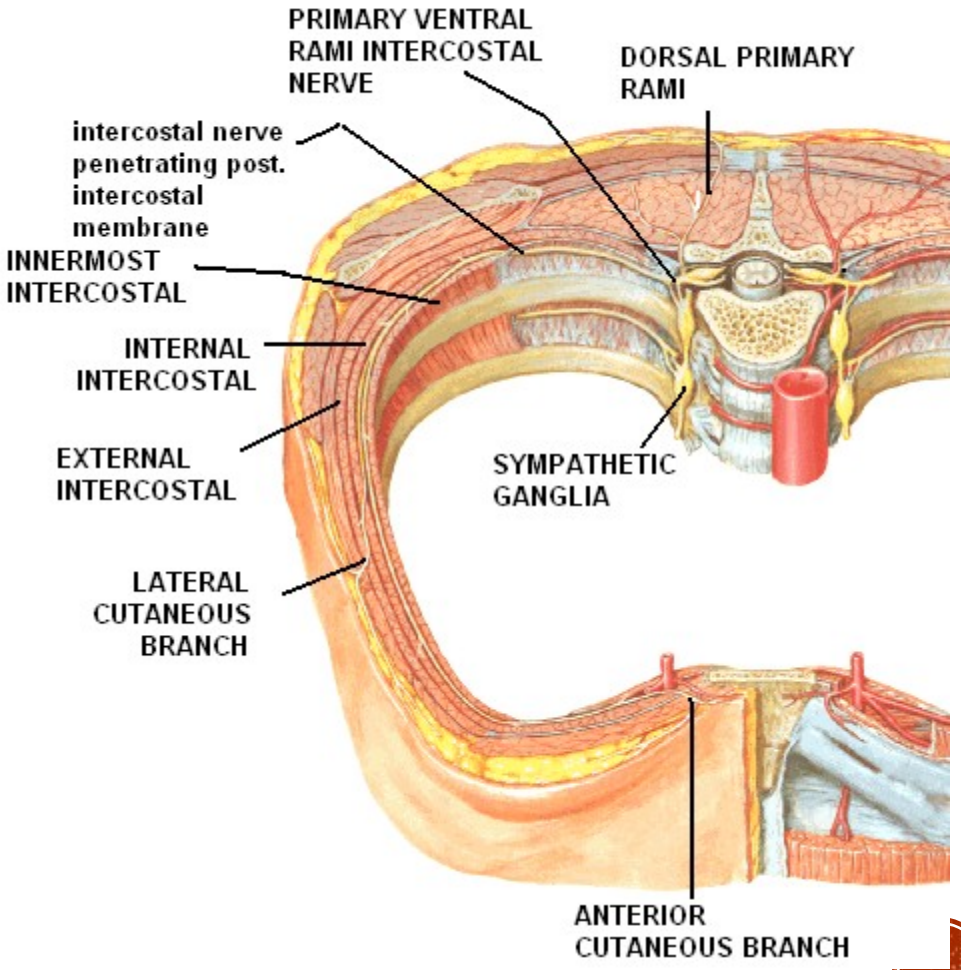
- **Posterior intercostal veins**
- •Drain into azygos or hemiazygos veins

- **Anterior intercostal veins**
- •Drain into internal thoracic and musculophrenic vein



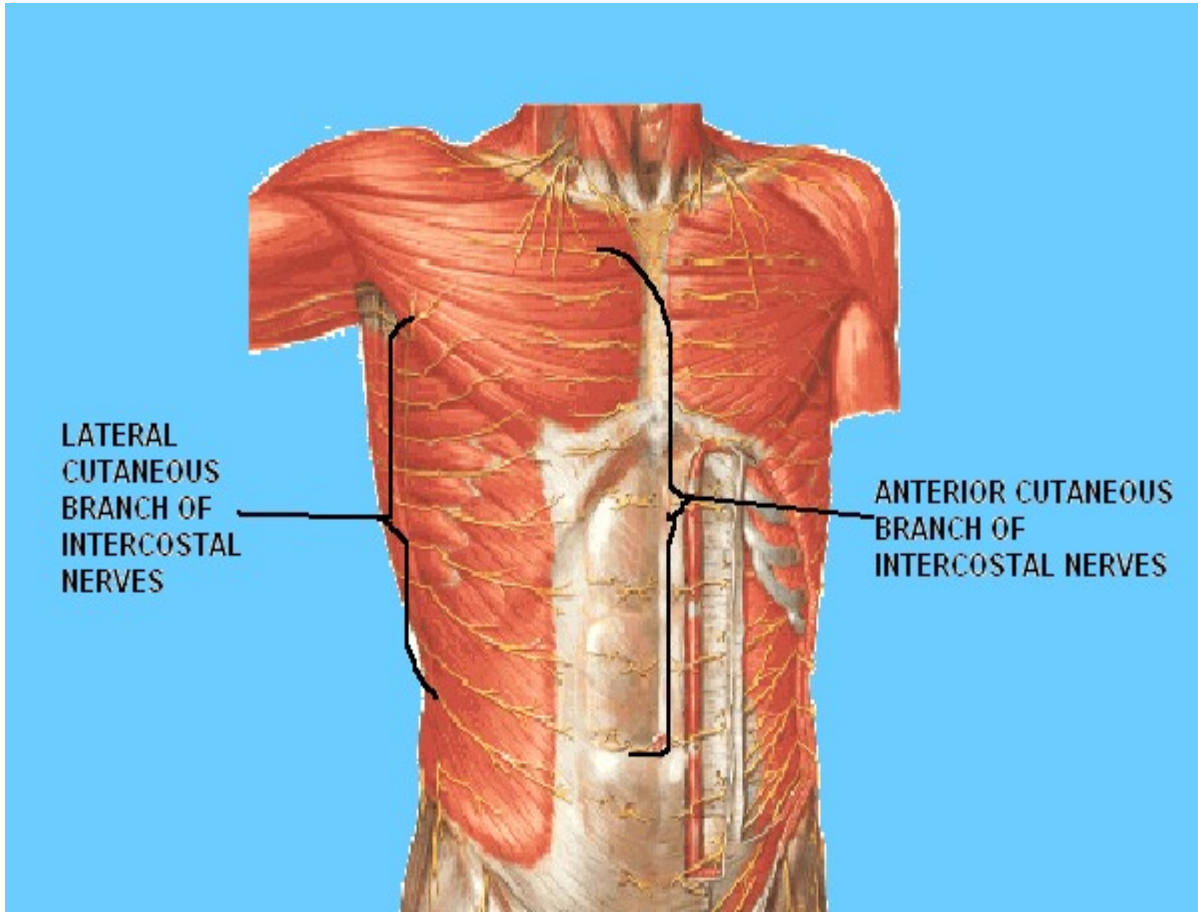
INTERCOSTAL NERVES

- Anterior primary rami of first 11 thoracic spinal nerves
- •Enter intercostal space between parietal pleura and posterior intercostal membrane
- •Runs at the lower border of rib between internal intercostal and inner most intercostal
- The 12th thoracic nerve lies in the abdomen and runs forward in the abdominal wall as subcostal nerve.



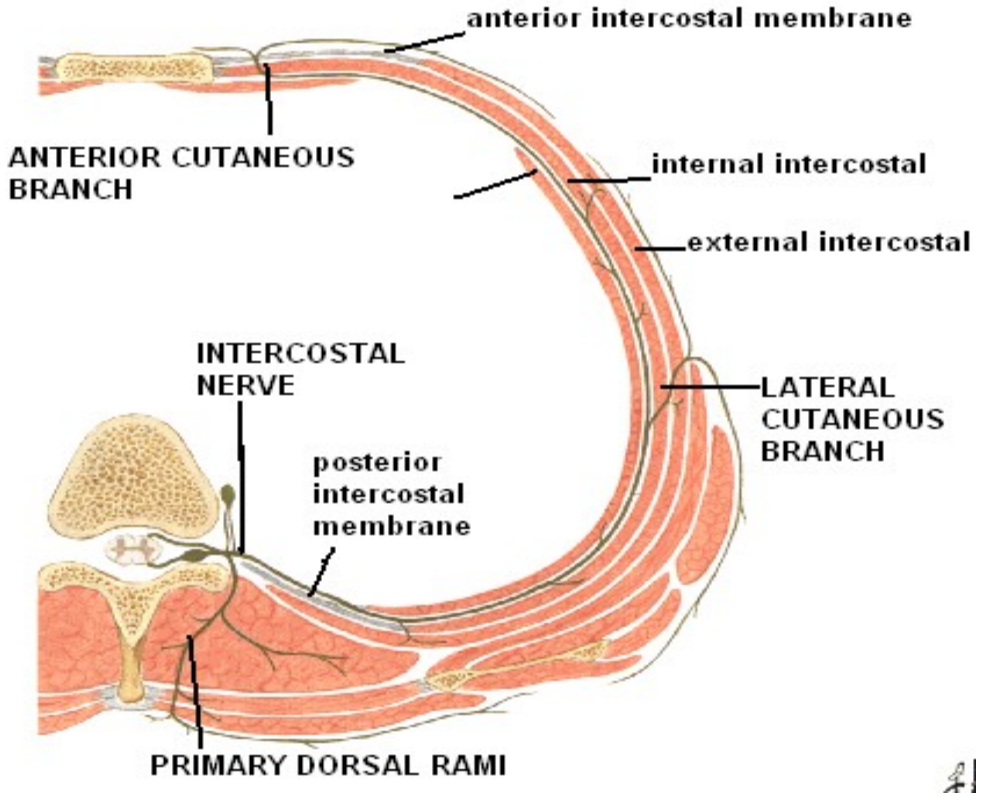
INTERCOSTAL NERVES

- 1st to 6th intercostal nerves remain with in their intercostal space
- 7th to 9th pass deep to the costal cartilage and then enter the abdominal wall
- 10th and 11th pass directly into the abdominal wall (since the corresponding ribs are floating ribs)



INTERCOSTAL NERVES

- **Branches**
- –Collateral branch runs in the same space at upper border of rib below
- –Lateral cutaneous branch >> skin on the sides of the chest
- –Anterior cutaneous branch >> skin on the midline
- –Muscular branches >> to the intercostal muscles
- –Pleural sensory branches >> to parietal pleura
- –Peritoneal sensory branches >> to parietal peritoneum

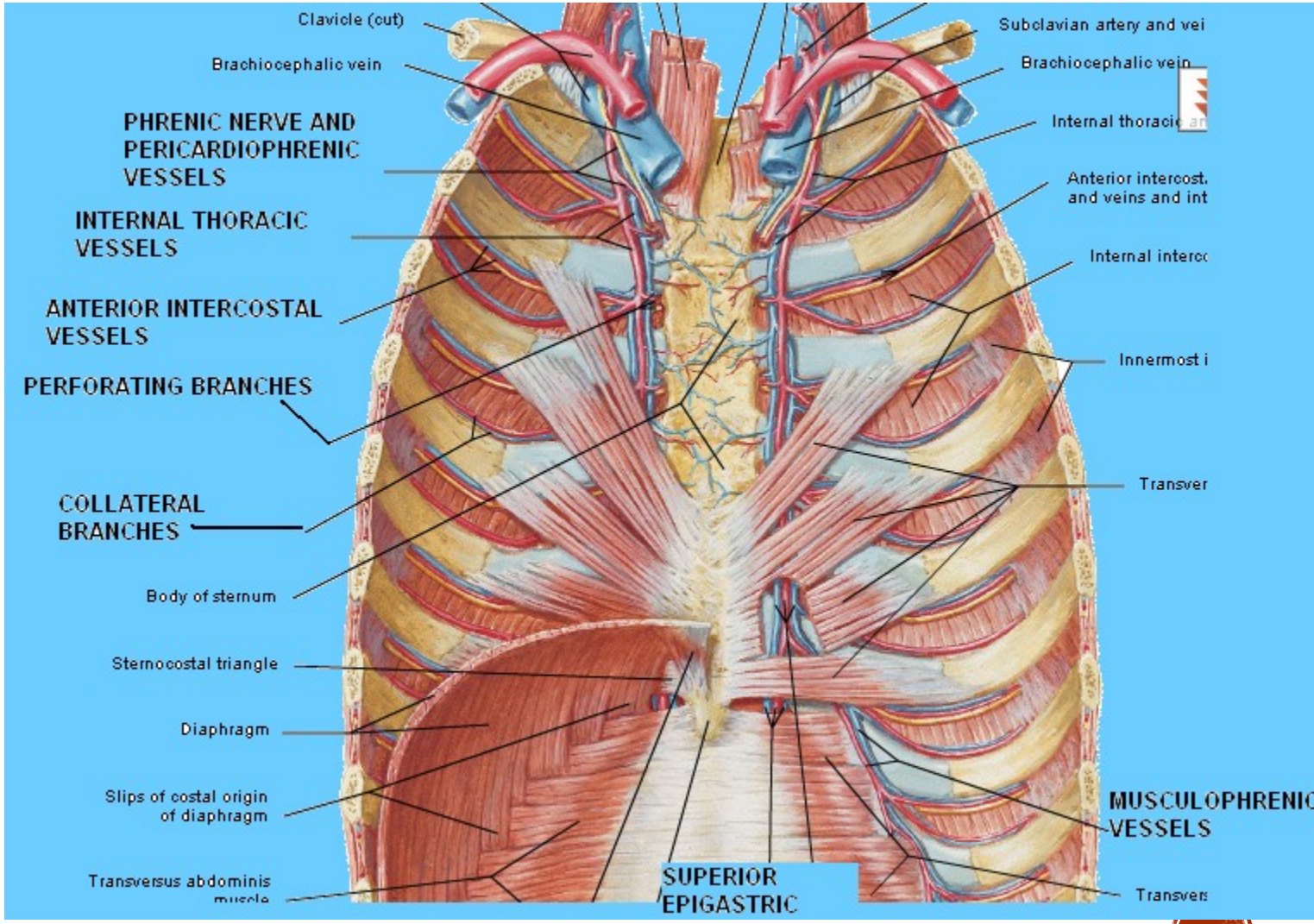


INTERCOSTAL NERVES

- **DISTRIBUTION**
- *1st to 6th supply:*
- •Skin and parietal pleura of intercostal spaces
- •Intercostal muscles of these spaces
- *7th to 11th supply*:
- •Skin and parietal peritoneum of outer inner surface of abdominal wall
- •Anterior abdominal wall muscles

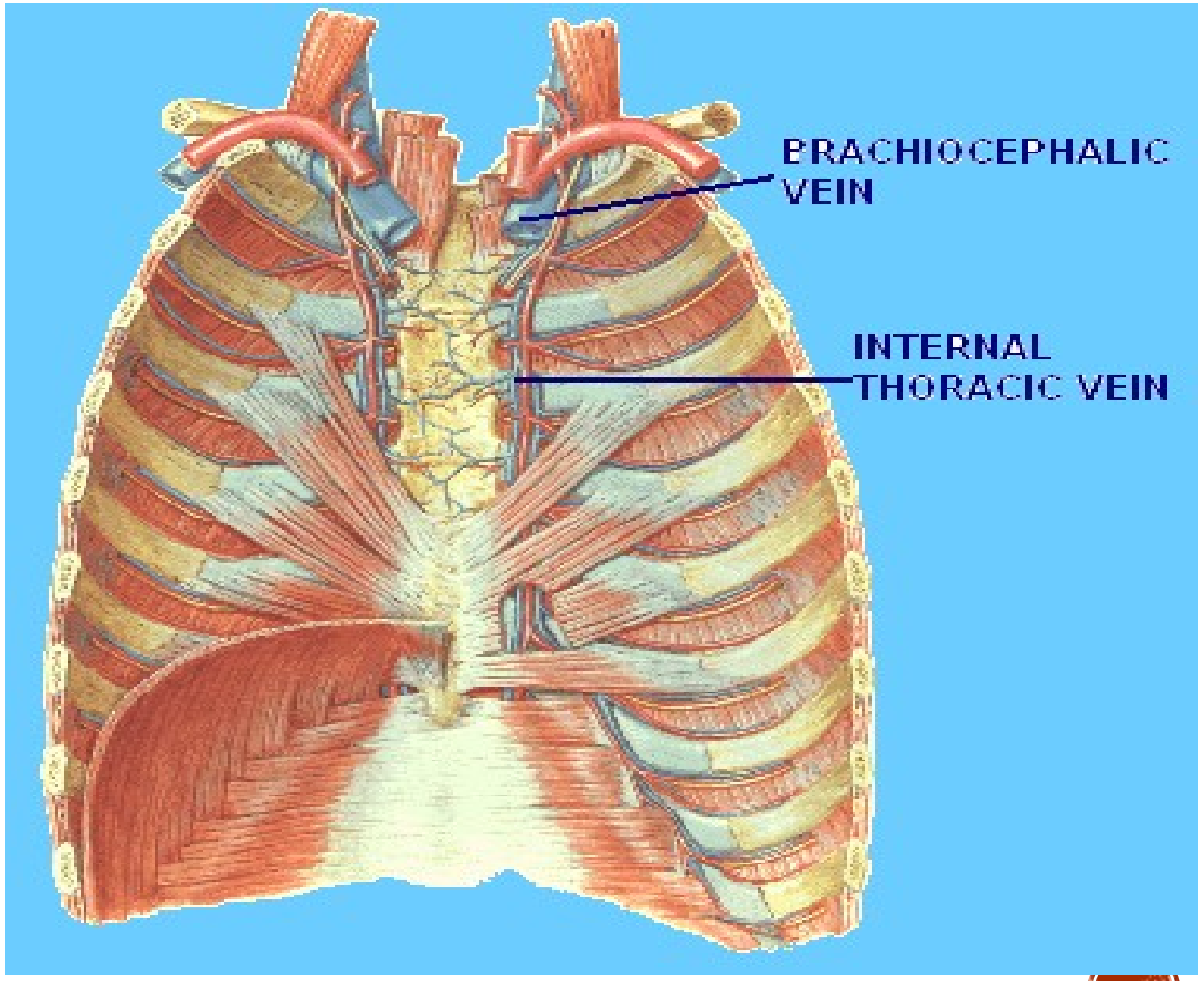
INTERNAL THORACIC ARTERY

- **Branches:**
- – 2 Anterior intercostal arteries for each of the upper six spaces
- –Perforating arteries
- –Pericardiophrenic artery
- –Mediastinal arteries
- –Superior epigastric artery
- –Musculophrenic artery



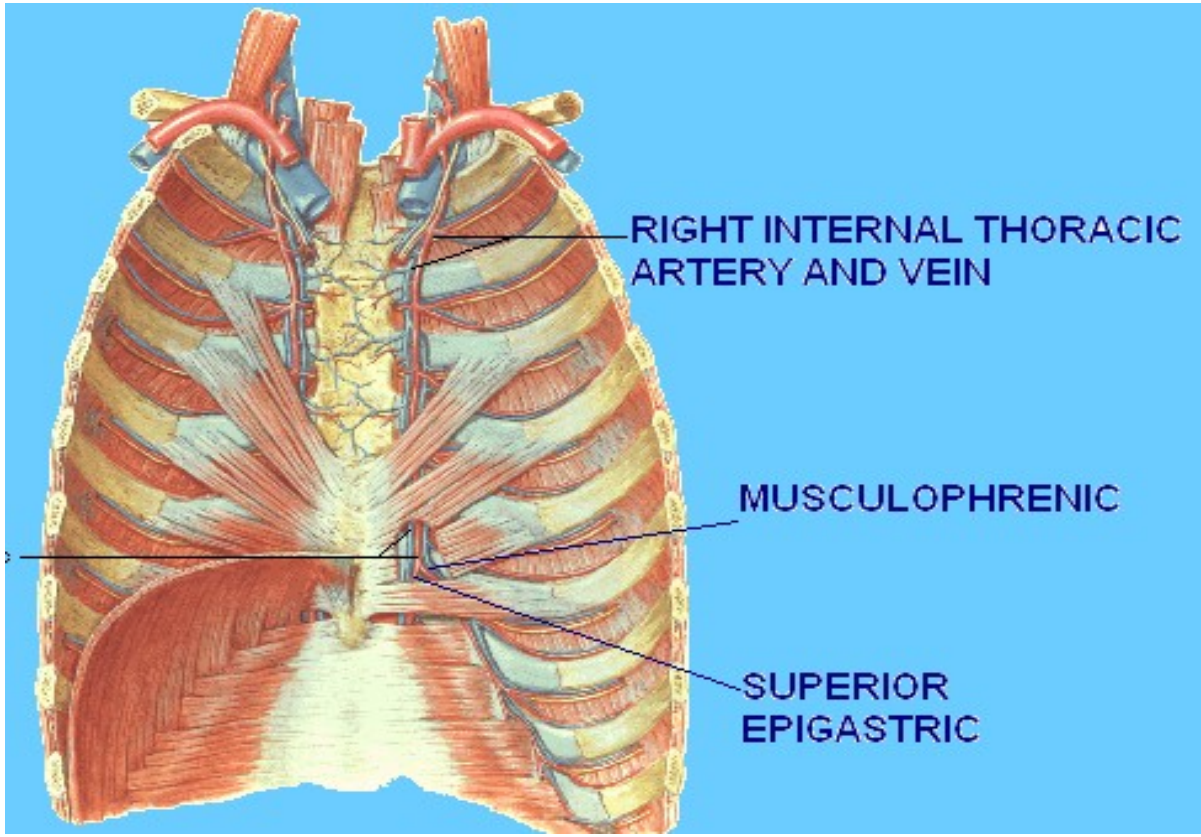
INTERNAL THORACIC VEIN

- Accompanies internal thoracic artery
- •Drains into brachiocephalic vein on each side

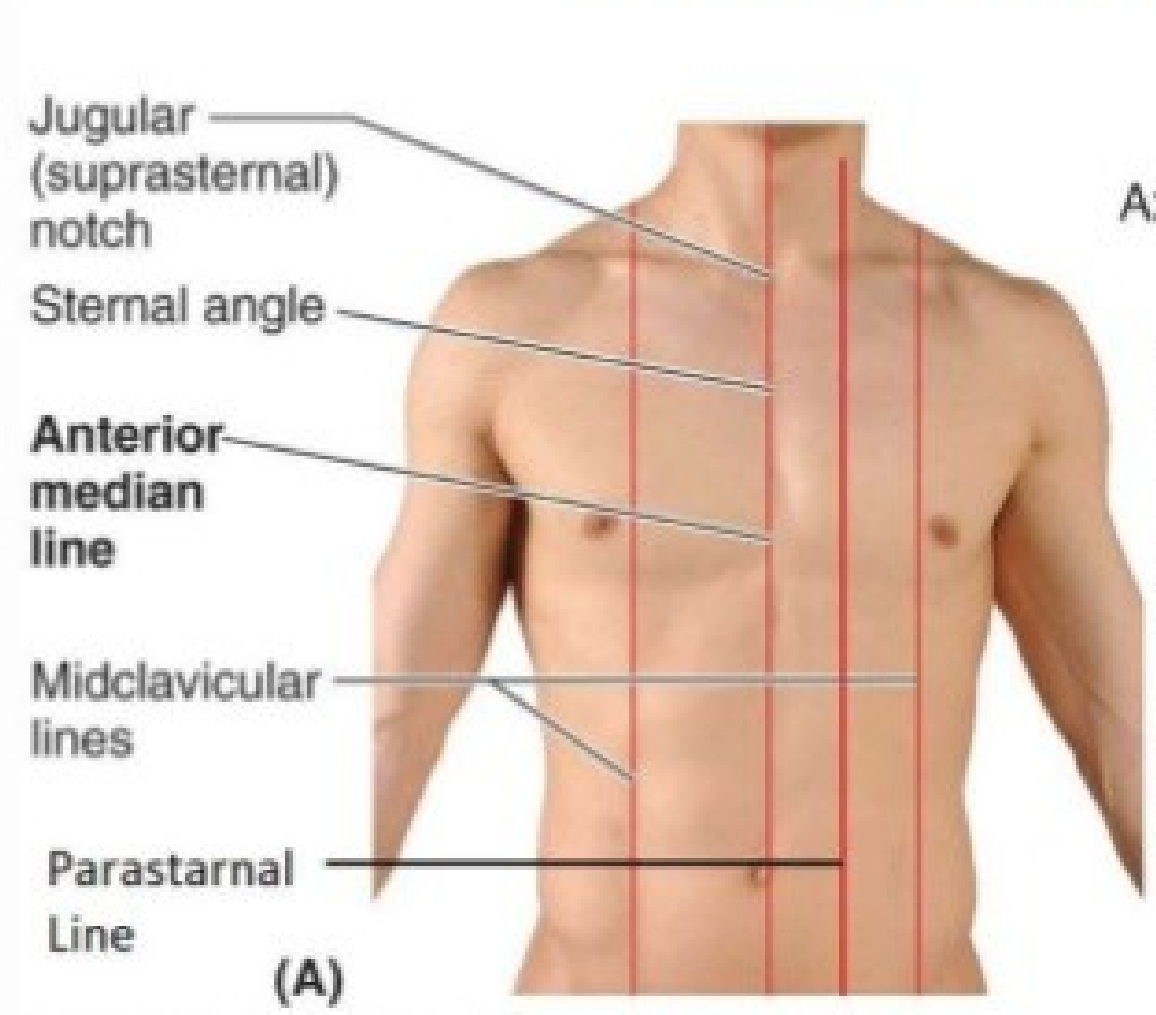


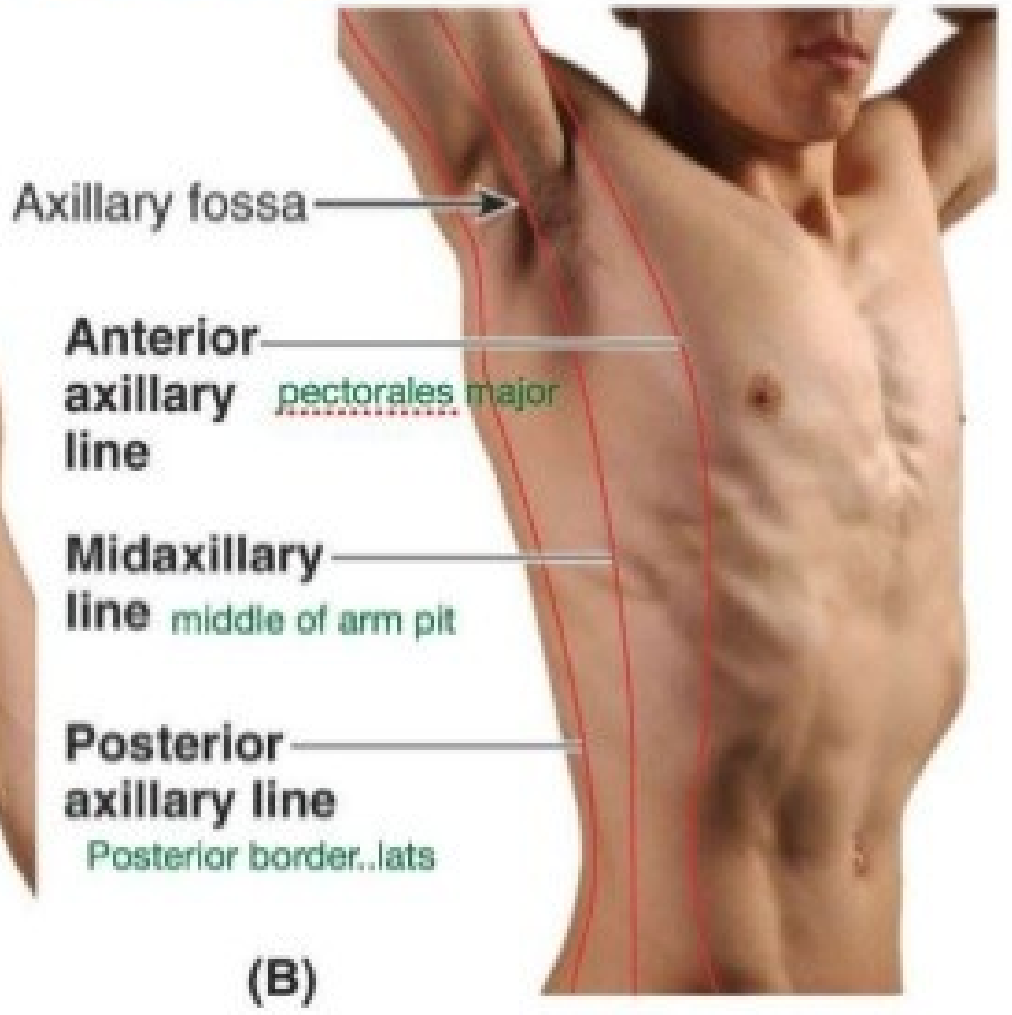
INTERNAL THORACIC ARTERY

- Branch of 1st part of subclavian artery in neck
- •Descends vertically on on the pleura behind costal cartilage
- •Ends in 6th intercostal space by dividing into ***superior epigastric and musculophrenic artery***



Different Lines





CLINICAL CORRELATES

Referred pain

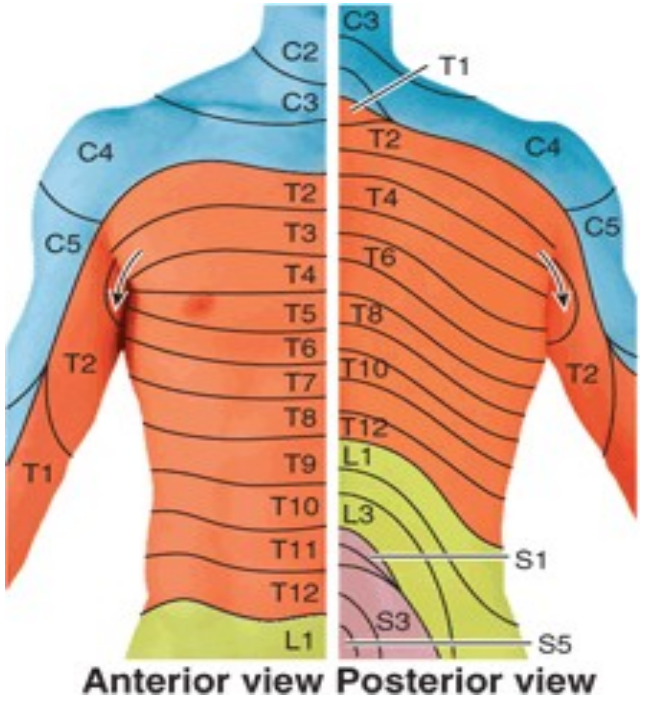
- If there is a Disease in thorax
- The Pain is felt along costal margin to anterior abdominal wall
- Because the 7th to 11th intercostal nerve supplies the anterior abdominal wall



CLINICAL CORRELATES

Herpes zoster

- It is caused by the reactivation of Latent varicella zoster virus in a patient who has previously had chickenpox.
- Mostly in patients above 50 years
- Inflammation and degeneration of sensory neuron
- Formation of vesicles with inflammation of skin
- In thorax, band of dermatomal pain in the distribution of the sensory neuron, followed by vesicles and disruption.





NEEDLE THORACOSTOMY

Insertion of needle in the pleural cavity

–Drainage of air, pus, blood

–Drawing a sample

•Anterior or lateral approach is used

Anterior approach: parient lying in supine position, 2nd intercostal space, mid clavicular line.

Lateral approach: patient lying on the lateral side, 2nd intercostal space, anterior axillary line

Needle passes through these structures:

1. **Skin,**

2. superficial fascia

3. Pectoral muscles (in anterior approach)

4. Serratus anterior

5. External intercostal muscles

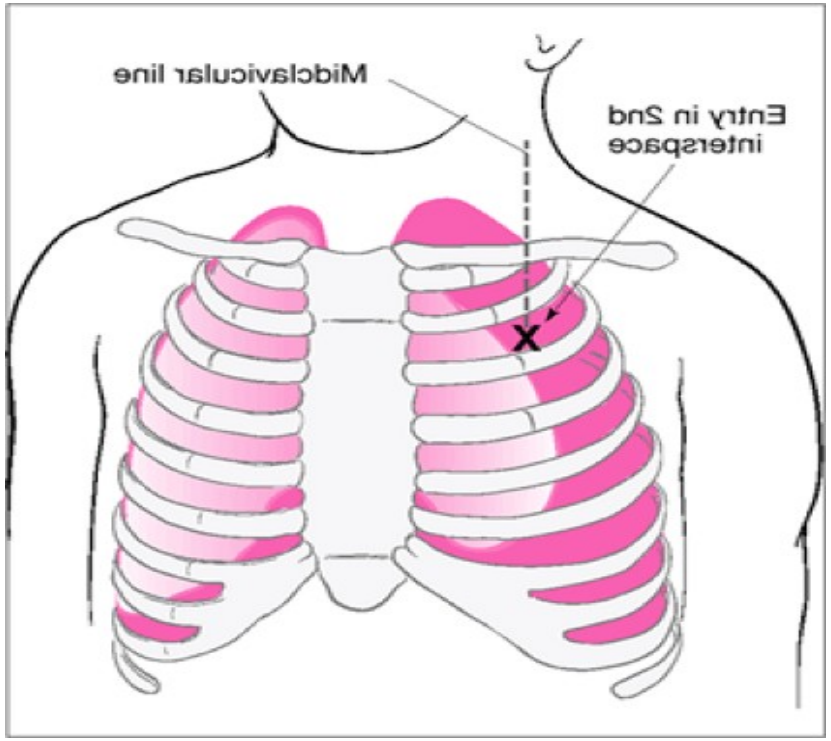
6. Internal intercostal muscles

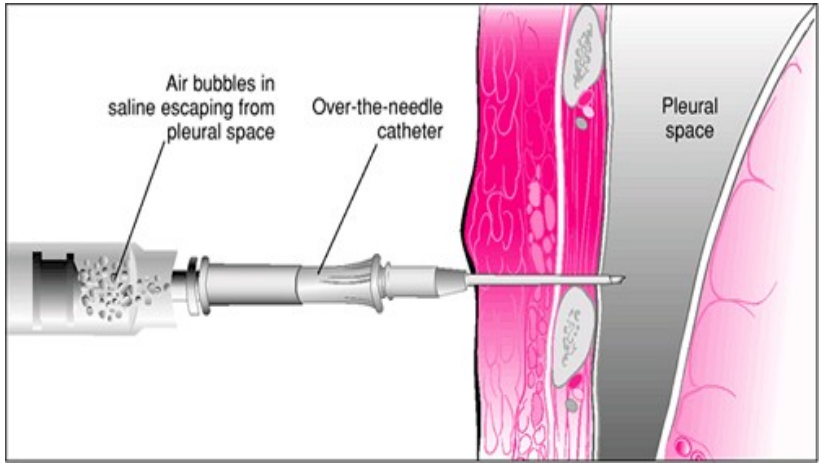
7. Inner most inter costal

8. Endothoracic fascia

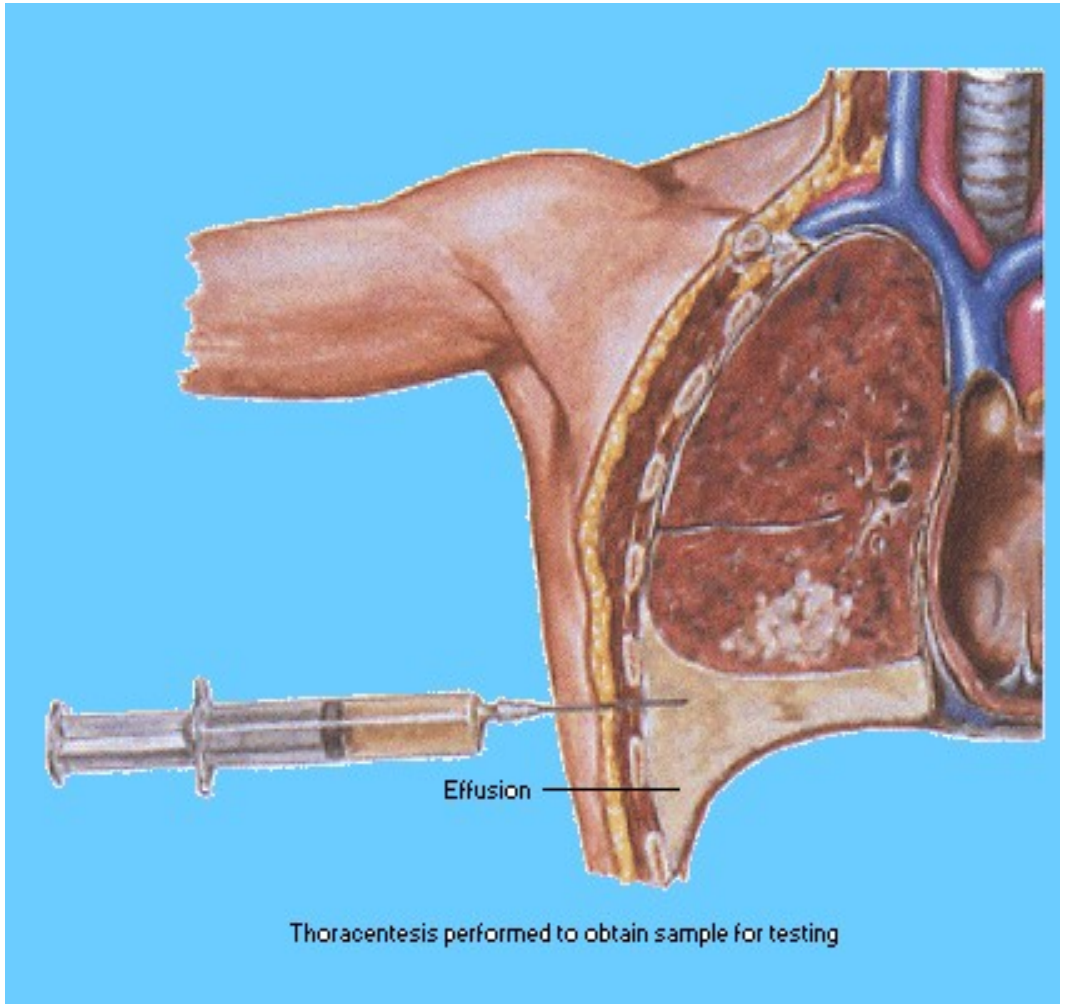
9. Parietal pleura

•Needle should be kept close to upper border of rib below to avoid injury to VAN

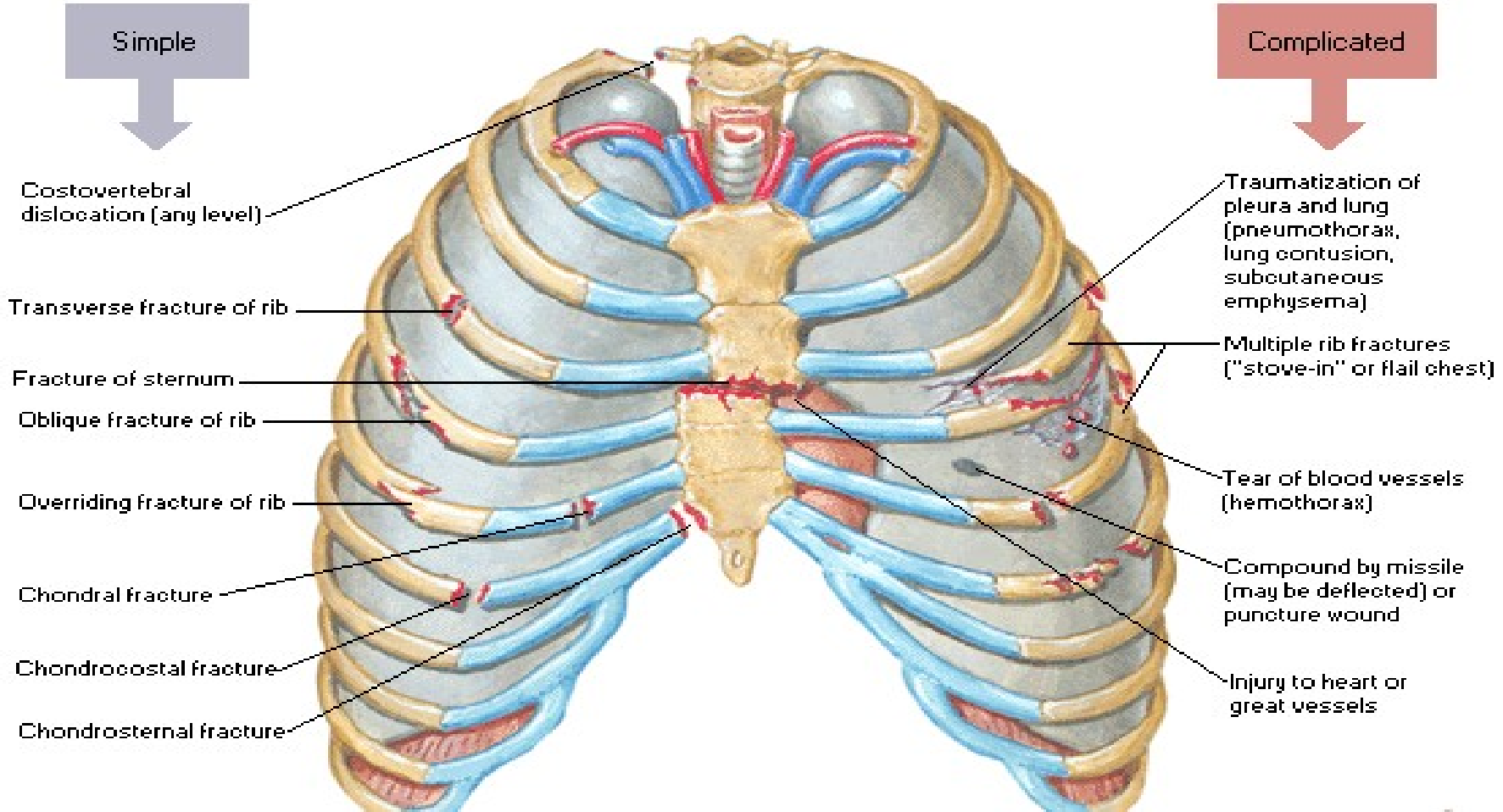




- Pleural fluid can be drained from lower intercoastal spaces , 4th or 5th intercostal space on anterior axillay line



INJURY TO THORACIC CAGE



Thank you



